

Principles of Helicopter Aerodynamics

J. GORDON LEISHMAN

University of Maryland



TO MY STUDENTS
in appreciation of all they have taught me

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Introduction: A History of Helicopter Flight

The idea of a vehicle that could lift itself vertically from the ground and hover motionless in the air was probably born at the same time that man first dreamed of flying.

Igor Ivanovitch Sikorsky

1.1 Introduction

The science of aerodynamics is the fundament of all flight. It is the role of aerodynamics in the engineering analysis and design of rotating-wing vertical lift aircraft that is the subject of this book. Igor Sikorsky's vision of a rotating-wing aircraft that could safely hover and perform other desirable flight maneuvers under full control of the pilot was only to be achieved some thirty years after fixed-wing aircraft (airplanes) were flying successfully. This rotating-wing aircraft we know today as the helicopter. Although the helicopter is considered by some to be a basic and somewhat cumbersome looking aircraft, the modern helicopter is indeed a machine of considerable engineering sophistication and refinement and plays a unique role in modern aviation provided by no other aircraft.

In the introduction to this book, the technical evolution of the helicopter is traced from a cumbersome, vibrating contraption that could barely lift its own weight into a modern and efficient aircraft that has become an indispensable part of modern life. Compared to fixed-wing flight, the development of which can be clearly traced to Lilienthal, Langley, and the first fully controlled flight of a piloted powered aircraft by the Wright Brothers in 1903, the origins of successful helicopter flight are less clear. Nonetheless, there are many parallels in the development of the helicopter when compared to fixed-wing aircraft. However, the longer and perhaps more tumultuous gestation period of the helicopter is directly attributable to the greater depth of scientific and aeronautical knowledge that was required before all the various technical problems could be understood and overcome. Besides the need to understand the basic aerodynamics of vertical flight and improve upon the aerodynamic efficiency of the helicopter, other technical barriers included the need to develop suitable high power-to-weight engines and high-strength, low-weight materials for the rotor blades, hub, fuselage, and transmission.

A helicopter can be defined as any flying machine using rotating wings (i.e., rotors) to provide lift, propulsion, and control forces that enable the aircraft to hover relative to the ground without forward flight speed to generate these forces. The thrust on the rotor(s) is generated by the aerodynamic lift forces created on the spinning blades. To turn the rotor, power from an engine must be transmitted to the rotor shaft. It is the relatively low amount of power required to lift the machine compared to other vertical take off and landing (VTOL) aircraft that makes the helicopter unique. Efficient hovering flight with low power requirements comes about by accelerating a large mass of air at a relatively low velocity; hence we have the large diameter rotors that are one obvious characteristic of helicopters. In addition, the helicopter must be able to fly forward, climb, cruise at speed, and then descend and come back into a hover for landing. This demanding flight capability comes at a price, including mechanical and aerodynamic complexity and higher power requirements

than for a fixed-wing aircraft of the same gross weight. All of these factors influence the design, acquisition, and operational costs of the helicopter.

Besides generating all of the vertical lift, the rotor is also the primary source of control and propulsion for the helicopter, whereas these functions are separated on a fixed-wing aircraft. For forward flight, the rotor disk plane must be tilted so that the rotor thrust vector is inclined forward to provide a propulsive component to overcome rotor and airframe drag. The orientation of the rotor disk to the flow also provides the forces and moments to control the attitude and position of the aircraft. The pilot controls the magnitude and direction of the rotor thrust vector by changing the blade pitch angles (using collective and cyclic pitch inputs), which changes the blade lift and the distribution of thrust over the rotor disk. By incorporating articulation into the rotor design through the use of mechanical flapping and lead/lag hinges that are situated near the root of each blade, the rotor disk can be tilted in any direction in response to these blade pitch inputs. As the helicopter begins to move into forward flight, the blades on the side of the rotor disk that advance into the relative wind will experience a higher dynamic pressure and lift than the blades on the retreating side of the disk, and so asymmetric aerodynamic forces and moments will be produced on the rotor. Articulation helps allow the blades to naturally flap and lag so as to help balance out these asymmetric aerodynamic effects. However, the mechanical complexity of the rotor hub required to allow for articulation and pitch control leads to high design and maintenance costs. With the inherently asymmetric flow environment and the flapping and pitching blades, the aerodynamics of the rotor become relatively complicated and lead to unsteady forces. These forces are transmitted from the rotor to the airframe and can be a source of vibrations, resulting in not only crew and passenger discomfort, but also considerably reduced airframe component lives and higher maintenance costs. However, with a thorough knowledge of the aerodynamics and careful design, all these adverse factors can be minimized or overcome to produce a highly reliable and versatile aircraft.

1.2 Early Attempts at Vertical Flight

There are many authoritative sources that record the development of helicopters and other rotating-wing aircraft such as autogiros. These include Gregory (1944), Lambermont (1958), Gablehouse (1967), Gunston (1983), Apostolo (1984), Boulet (1984), Lopez & Boyne (1984), Taylor (1984), Everett-Heath (1986), Fay (1987) and Spenser (1999), amongst others. Boulet (1984) takes a unique approach in that he gives a first-hand account of the early helicopter developments through interviews with the pioneers, constructors, and pilots of the machines. A remarkably detailed history of early helicopter developments is given by Liberatore (1950, 1988, 1998). For original publications documenting early technical developments of the autogiro and helicopter, see Warner (1920), von Kármán (1921), Balaban (1923), Moreno-Caracciolo (1923), Klemm (1925), Wimperi (1926), and Seiferth (1927).

As described by Liberatore (1998), the early work on the development of the helicopter can be placed into two distinct categories: inventive and scientific. The former is one where intuition is used in lieu of formal technical training, whereas the latter is one where a trained, systematic approach is used. Prior to the nineteenth century there were few scientific investigations of flight or the science of aerodynamics. The inherent mechanical and aerodynamic complexities in building a practical helicopter that had adequate power and control, and did not vibrate itself to pieces, resisted many ambitious efforts. The history of flight documents literally hundreds of failed helicopter projects, which, at most, made

only brief uncontrolled hops into the air. Some designs provided a contribution to new knowledge that ultimately led to the successful development of the modern helicopter. Yet, it was not until the more scientific contributions of engineers such as Juan de la Cierva, Heinrich Focke, Raoul Hafner, Igor Sikorsky, Arthur Young, and others did the design of a truly safe and practical helicopter become a reality.

Six fundamental technical problems can be identified that limited early experiments with helicopters. These problems are expounded by Sikorsky (1938, and various editions) in his autobiography. In summary, these problems were:

1. Understanding the aerodynamics of vertical flight. The theoretical power required to produce a fixed amount of lift was an unknown quantity to the earliest experimenters, who were guided more by intuition than by science.¹
2. The lack of a suitable engine. This was a problem that was not to be overcome until the beginning of the twentieth century, through the development of internal combustion engines.
3. Keeping structural weight and engine weight down so the machine could lift a pilot and a payload. Early power plants were made of cast iron and were heavy.²
4. Counteracting rotor torque reaction. A tail rotor was not used on most early designs; these machines were either coaxial or laterally side-by-side rotor configurations. Yet, building and controlling two rotors was even more difficult than for one rotor.
5. Providing stability and properly controlling the machine, including a means of defeating the unequal lift produced on the advancing and retreating blades in forward flight. These were problems that were only to be fully overcome with the use of blade articulation, ideas that were pioneered by Cierva, Breguet, and others, and with the development of blade cyclic pitch control.
6. Conquering the problem of vibrations. This was a source of many mechanical failures of the rotor and airframe, because of an insufficient understanding of the dynamic and aerodynamic behavior of rotating wings.

The relatively high weight of the structure, engine, and transmission was mainly responsible for the painfully slow development of the helicopter until about 1920. However, by then gasoline powered piston engines with higher power-to-weight ratios were more widely available, and the antitorque and control problems of achieving successful vertical flight were at the forefront. This resulted in the development of a vast number of prototype helicopters. Many of the early designs were built in Great Britain, France, Germany, Italy, and the United States, who led the field in several technical areas. However, with all the various incremental improvements that had been made to the basic helicopter concept during the pre-World War II years, it was not until the late inter war period that significant technical advances were made and more practical helicopter designs began to appear. The most important advances of all were in engine technology, both piston and gas turbines, the latter of which revolutionized both fixed-wing and rotating-wing flight.

A time-line documenting the evolution of rotating-wing aircraft through 1950 is shown in Fig. 1.1. The ideas of vertical flight can be traced back to early Chinese tops, a toy first used about 400 BC. Everett-Heath (1986) and Liberatore (1998) give a detailed history of such devices. The earliest versions of the Chinese top consisted of feathers at the end of

¹ The first significant application of aerodynamic theory to helicopter rotors came about in the early 1920s.

² Aluminum was not available commercially until about 1890 and was inordinately expensive. It was not used in aeronautical applications until about 1915.

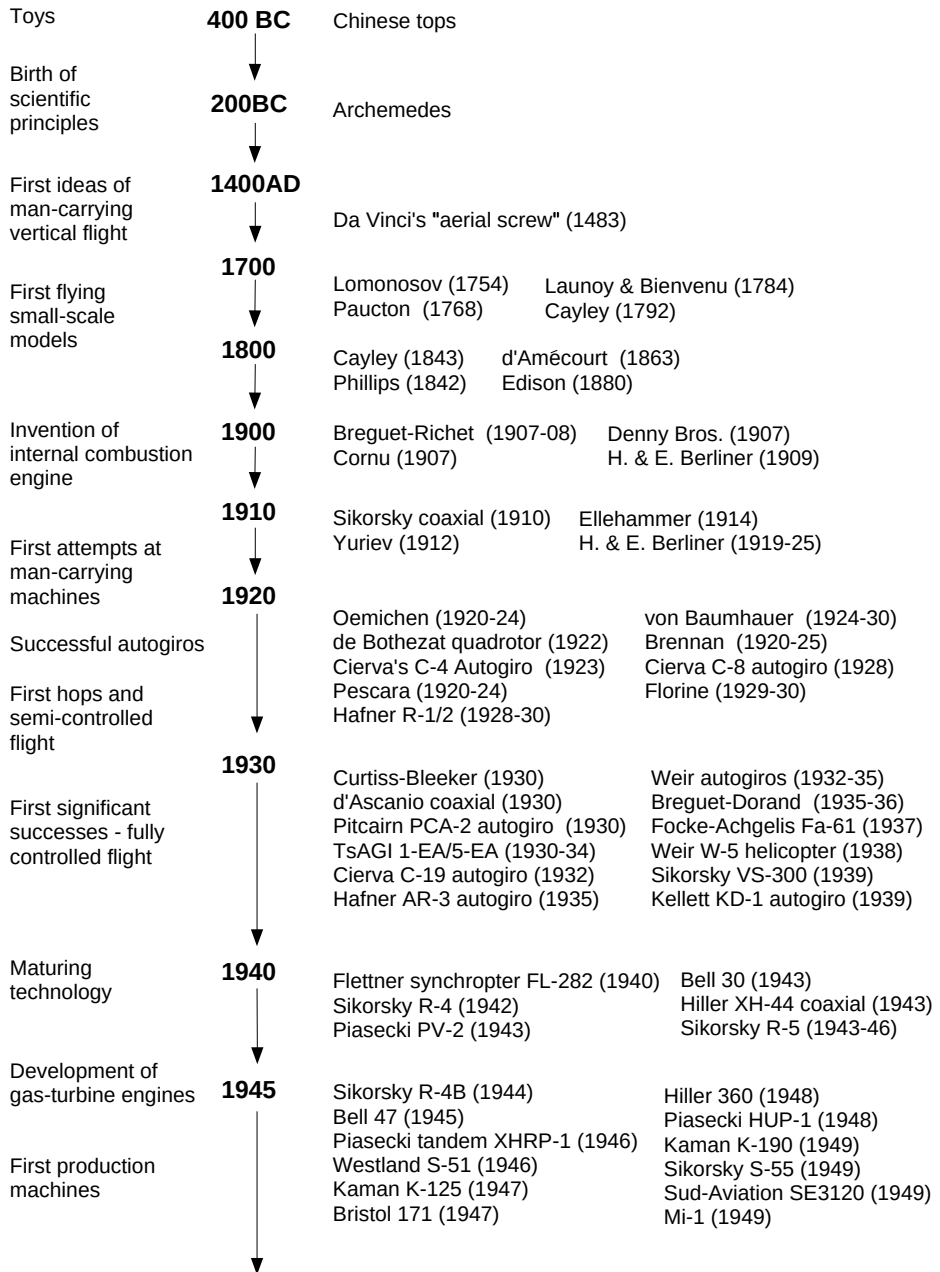


Figure 1.1 Time-line showing development of helicopters and autogiros prior to 1950.

a stick, which was rapidly spun between the hands to generate lift and then released into flight. More than 2,000 years later in 1784, Launoy & Bienvenu used a coaxial version of the Chinese top in a model consisting of a counterrotating set of turkey feathers, powered by a string wound around its shaft and tensioned by a crossbow. It is also recorded that Mikhail Lomonosov of Russia had developed, as early as 1754, a small coaxial rotor modeled after the Chinese top but powered by a wound-up spring device. In 1786, the French mathematician

A. J. P. Paucton published a paper entitled “Théorie de la vis D’Archimède,” where he proposed a human-carrying flying machine, with one rotor to provide lift and another for propulsion.

Amongst his many intricate drawings, Leonardo da Vinci shows what is a basic human-carrying helicopterlike machine, an obvious elaboration of an Archimedes water-screw. His sketch of the “aerial-screw” device, which is shown in Fig. 1.2, is dated to 1483 but was first published nearly three centuries later. The device comprises a helical surface that da Vinci describes should be “rotated with speed that said screw bores through the air and climbs high.” He realized that the density of air is much less than that of water, and so da Vinci describes how the device needed to be relatively large to accomplish this feat (the number “8” in his writing to the left of the sketch indicates that the size of the rotor is 8 *braccia* or arm lengths). He also describes in some detail how the machine should be built using wood, wire, and linen cloth. Although da Vinci worked on various concepts of

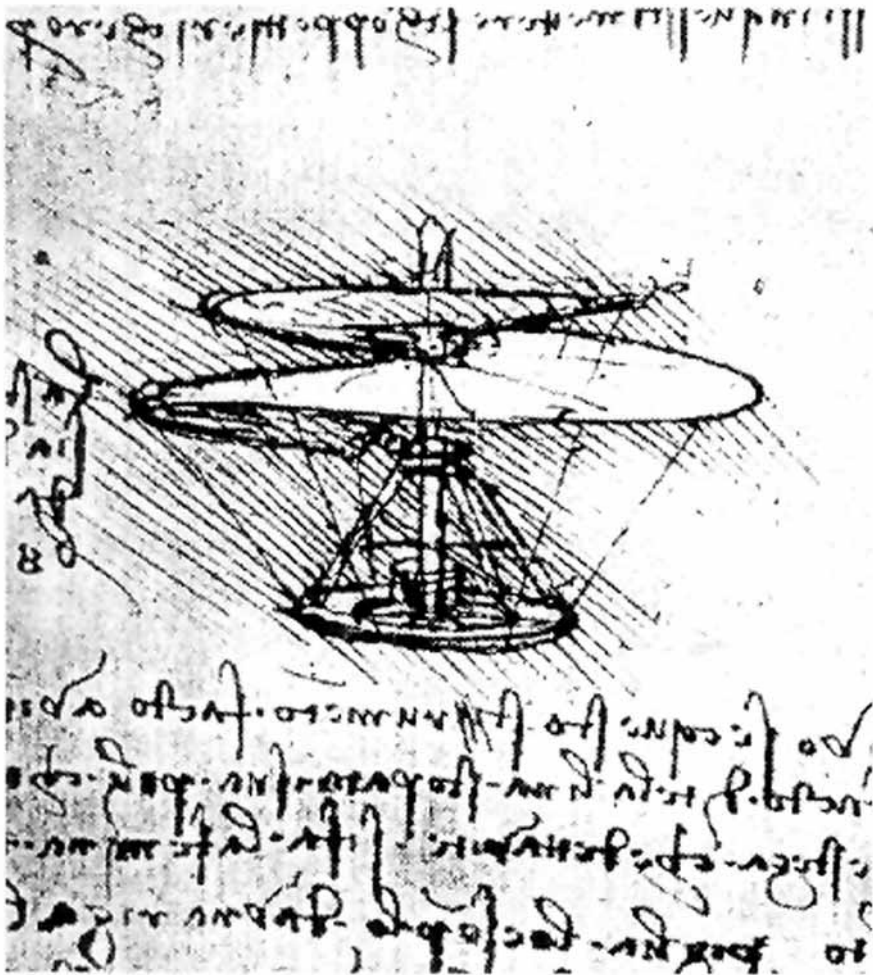


Figure 1.2 Leonardo da Vinci's aerial screw machine, dated to 1483. Original drawing is MS 2173 of Manuscript (codex) B, folio 83 verso, in the collection of the Bibliothèque L'Institut de France (Paris).

engines, turbines, and gears, he did not unite the ideas of his aerial-screw machine to an engine nor did he appreciate the problems of torque reaction. See Hart (1961) or Giacomelli (1930) for further details of da Vinci's aeronautical work.

Sir George Cayley is famous for his work on the basic principles of flight, which dates from the 1790s – see Pritchard (1961). By the end of the eighteenth century, Cayley had constructed several successful vertical-flight models based on Chinese tops driven by wound-up clock springs. He designed and constructed a whirling-arm device in 1804, which was probably one of the first scientific attempts to study the aerodynamic forces produced by lifting wings. Cayley (1809–10) published a three-part paper that was to lay down the foundations of aerodynamics – see Anderson (1997). In a later paper, published in 1843, Cayley gives details of a vertical flight aircraft design that he called an “Aerial Carriage,” which had two pairs of lateral side-by-side rotors. Also, in the 1840s, another Englishman, Horatio Phillips, constructed a steam-driven vertical flight machine, where steam generated by a miniature boiler was ejected out of the blade tips. Although impractical, Phillips's machine was significant in that it marked the first time that a model helicopter had flown under the power of an engine rather than stored energy devices such as wound-up springs.

In the early 1860s, Ponton d'Amécourt of France flew a number of small helicopter models. He called his machines *hélicoptères*, which is a word derived from the Greek adjective *elikoeioas*, meaning spiral or winding, and the noun *pteron*, meaning feather or wing – see Wolf (1974) and Liberatore (1998). In 1863, d'Amécourt built a steam propelled model helicopter, but it could not generate enough lift to fly. However, the novelist Jules Verne was still impressed with d'Amécourt's attempts, and in 1886 he wrote “The Clipper of the Clouds” where the hero cruised around the skies in a giant helicopterlike machine that was lifted by thirty-seven small coaxial rotors and pulled through the air by two propellers.

Other notable vertical flight models that were constructed at about this time include the coaxial design of Bright in 1861 and the twin-rotor steam-driven model of Dieuaide in 1877. Wilhelm von Achenbach of Germany built a single rotor model in 1874, and he was probably the first to use the idea of a tail rotor to counteract the torque reaction from the main rotor. Later, Achenbach conducted experiments with propellers, the results of which were published by NACA – see Achenbach (1923). About 1869 a Russian helicopter concept was developed by Lodygin, using a rotor for lift and a propeller for propulsion and control. Around 1878, Enrico Forlanini of Italy also built a flying steam-driven helicopter model. This model had dual counterrotating rotors, but like many other model helicopters of the time, it was underpowered and had no means of control.

In the 1880s, Thomas Alva Edison experimented with small helicopter models in the United States. He tested several rotor configurations driven by a guncotton engine, which was an early form of internal combustion engine. Later, Edison used an electric motor for power, and he was one of the first to realize from his experiments the need for a large diameter rotor with low solidity to give good hovering efficiency [Liberatore (1998)]. Unlike other experimenters of the time, Edison's more scientific approach to the problem proved that both high aerodynamic efficiency of the rotor and high power from an engine were required if successful vertical flight was to be achieved. In 1910, Edison patented a rather cumbersome looking full-scale helicopter concept with boxkite-like blades, but there is no record that it was ever constructed.

In 1907, about four years after the Wright brothers' first successful powered flights in fixed-wing airplanes at Kitty Hawk in the United States, Paul Cornu of France constructed a vertical flight machine that carried a human off the ground for the first time. Boulet (1984) gives a good account of the work. The airframe was very simple, with a rotor at each end

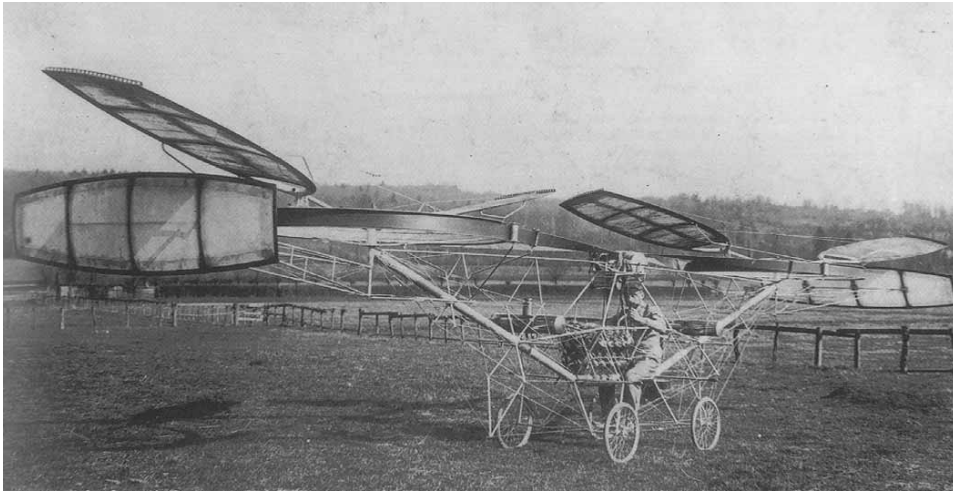


Figure 1.3 The Cornu helicopter, circa 1907. (Courtesy NASM, Smithsonian Institution, SI Neg. No. 74-8533.)

(Fig. 1.3). Power was supplied to the rotors by a gasoline motor and belt transmission. Each rotor had two blades, and the rotors rotated in opposite directions to cancel torque reaction. A primitive means of control was achieved by placing a wing in the slipstream below the rotors. The machine was reported to have made several tethered flights of a few seconds at low altitude. Also in France, the Breguet brothers had begun to conduct helicopter experiments about 1907. Their complicated quadrotor “Gyroplane” carried a pilot off the ground, albeit briefly, but like the Cornu machine it was underpowered, and it lacked stability and a proper means of control.

In the early 1900s, Igor Sikorsky and Boris Yur’ev independently began to design and build vertical-lift machines in Czarist Russia. By 1909, Sikorsky had built a nonpiloted coaxial prototype. This machine did not fly because of vibration problems and the lack of a powerful enough engine. Sikorsky (1938) stated that he had to await “better engines, lighter materials, and experienced mechanics.” His first design was unable to lift its own weight, and the second, even with a more powerful engine, only made short (nonpiloted) hops. Sikorsky abandoned the helicopter idea and devoted his skills to fixed-wing (conventional airplane) designs at which he was very successful. Although he never gave up his vision of the helicopter, it was not until the 1930s after he had emigrated to the United States that he pursued his ideas again (see Section 1.4). Good accounts of the life and work of Igor Sikorsky are documented by Bartlett (1947), Delear (1969), Sikorsky (1964, 1971), Sikorsky & Andrews (1984), Finne (1987), and Cochrane et al. (1989).

Unbeknown to Sikorsky, Boris Yur’ev had also tried to build a helicopter in Russia around 1912, but with a single rotor and tail rotor configuration. Like Sikorsky’s machine, the aircraft lacked a powerful enough engine. Besides being one of the first to use a tail rotor design, Yur’ev was one of the first to propose the concept of cyclic pitch for rotor control. (Another early design was patented by Gaetano Crocco of Italy in 1906). Good accounts of Yur’ev’s machine are given by Gablehouse (1967) and Liberatore (1998). There is also evidence of the construction of a primitive coaxial helicopter by Professor Zhukovskii (Joukowski) and his students at Moscow University in 1910 – see Gablehouse (1967). Joukowski is well known for his theoretical contributions to aerodynamics, and he published several papers on the subject of rotating wings and helicopters; see also Margoulis (1922) and Tokaty (1971).



Figure 1.4 Danish aviation pioneer Jens Ellehammer flew a coaxial rotor helicopter design in 1914.

About 1914, the Danish aviation pioneer Jens Ellehammer designed a coaxial rotor helicopter. Boulet (1984) gives a good description of the machine, which is shown in Fig. 1.4. The rotor blades themselves were very short; six of these were attached to each of two large circular aluminum rings. The lower disk was covered with fabric and was intended to serve as a parachute in the event the blades or engine failed. A cyclic pitch mechanism was used to provide control, this being another one of many early applications of the concept. The pilot was supported in a seat that could be moved forward and sideways below the rotor, allowing for additional kinesthetic control. The aircraft made many short hops into the air but never made a properly controlled free flight.

An Austrian, Stephan Petroczy, with the assistance of the well-known aerodynamicist Theodore von Kármán, built and flew a coaxial rotor helicopter during 1917–1920. Interesting design features of this machine included a pilot/observer position above the rotors, inflated bags for landing gear, and a quick-opening parachute. While the machine never really flew freely, it accomplished numerous limited tethered vertical flights. The work is summarized in a report by von Kármán (1921) and published by the NACA. It is significant that von Kármán also gives results of laboratory tests on the “rotors,” which were really oversize propellers. With the work of William F. Durand [see Warner (1920) and the analysis by Munk (1923)] these were some of the first attempts to scientifically study rotor performance and the power required for vertical flight.

In the United States, Emile and Henry Berliner (a father and son) were interested in vertical flight aircraft. As early as 1909, they had designed and built a helicopter based on pioneering forward flight experiments with a wheeled test rig. In 1918 the Berliners patented a single-rotor helicopter design, but there is no record that this machine was built. Instead, by about 1919, Henry Berliner had built a counterrotating coaxial rotor machine, which made brief uncontrolled hops into the air. By the early 1920s at the College



Figure 1.5 This Berliner helicopter with side-by-side rotors made short flights at College Park airport in Maryland in 1922. (Courtesy of College Park Airport Museum.)

Park airport, the Berliners were flying an aircraft with side-by-side rotors (Fig. 1.5). The rotors were oversized wooden propellers, but with special airfoil profiles and twist distributions. Differential longitudinal tilt of the rotor shafts provided yaw control. On later variants, lateral control was aided by cascades of wings located in the slipstream of the rotors. All variants used a conventional elevator and rudder assembly at the tail, also with a small vertically thrusting auxiliary rotor on the rear of the fuselage. The Berliner's early flights with the coaxial rotor and side-by-side rotor machines are credited as some of the first rudimentary piloted helicopter developments in the United States. However, because true vertical flight capability with these machines was limited, the Berliners abandoned the pure helicopter in favor of a hybrid machine they called a "helicoplane." This still used the rotors for vertical lift but incorporated a set of triplane wings and a larger oversized rudder. The Berliner's final hybrid machine of 1924 was a biplane wing configuration with side-by-side rotors. See also Berliner (1908, 1915).

In Britain during the 1920s, Louis Brennan worked on a helicopter concept with an unusually large single two-bladed rotor. Fay (1987) gives a good account of Brennan's work. Brennan, who was an inventor of some notoriety, had a different approach to solving the problem of torque reaction by powering the rotor with propellers mounted on the blades (Fig. 1.6). Control was achieved by the use of "ailerons" inboard of the propellers. In 1922, the machine lifted off inside a balloon shed. Further brief low altitude flights outdoors were undertaken through 1925, but the machine crashed, and further work stopped because of increasing interest in the autogiro (see Section 1.3).

During the early 1920s, Raul Pescara, an Argentinean living and working in Spain and France, was building and attempting to fly a coaxial helicopter with biplane-type rotors (Fig. 1.7). As described by Boulet (1984), each rotor had a remarkable five sets of biplane blades that were mounted rigidly to the rotor shaft. Pescara's work focused on the need

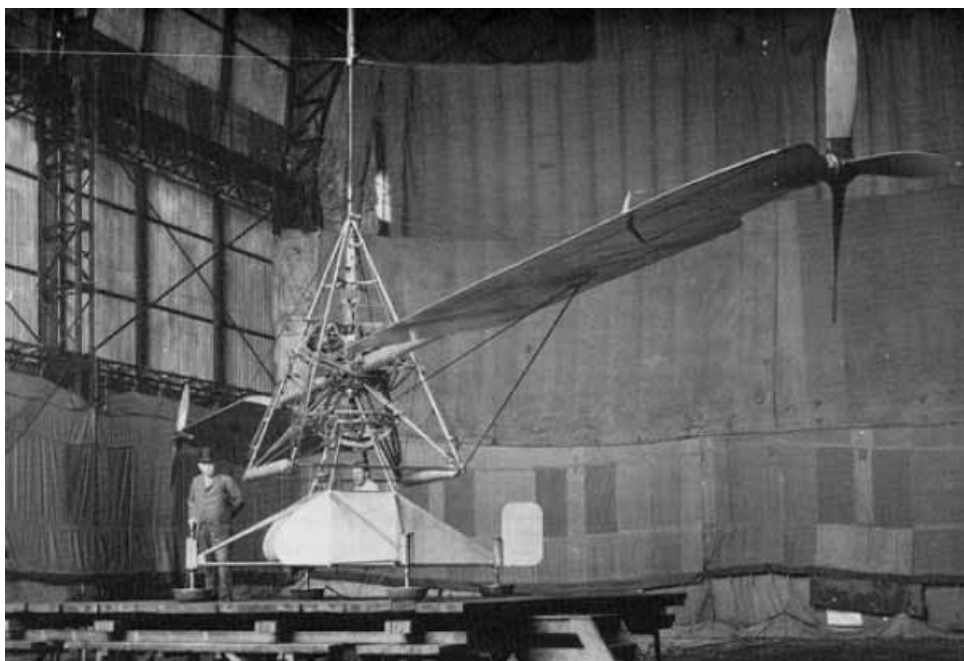


Figure 1.6 The Brennan helicopter suspended in the balloon shed at RAE Farnborough, circa 1922.

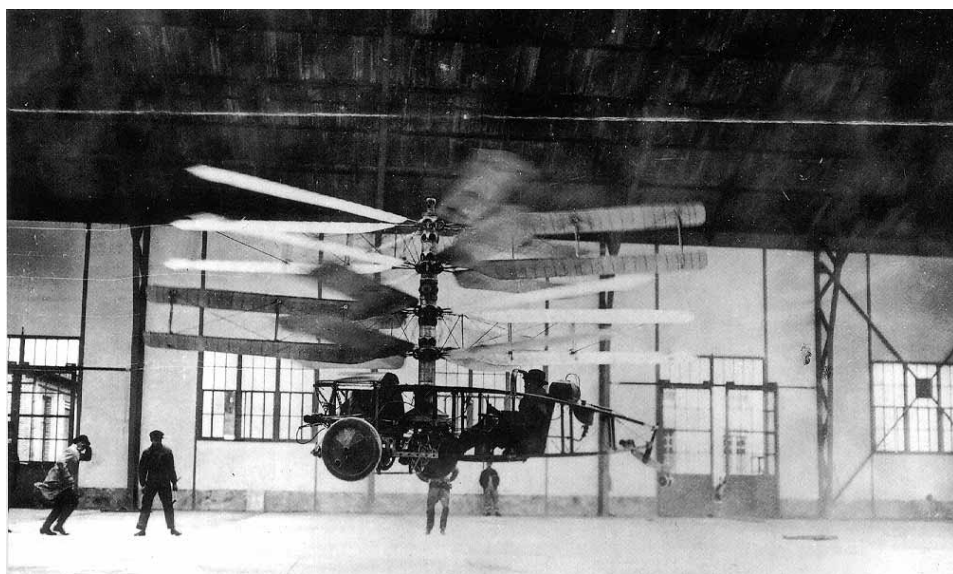


Figure 1.7 Pescara's helicopter hovering in a hanger about 1923. (Courtesy NASM Smithsonian Institution, SI Neg. No. 83-16343.)



Figure 1.8 Between 1924 and 1930, A. G. von Baumhauer made attempts to fly a single main rotor helicopter with a separately powered tail rotor. (Courtesy of NASM, Smithsonian Institution, Neg. No. 77-721.)

for complete control of the machine, which was achieved through cyclic-pitch changes that could be obtained by warping the blades periodically as they rotated. This was one of the first successful applications of cyclic pitch. Yaw was controlled by differential collective pitch between the two rotors. Early versions of his machine were underpowered, which may not be surprising considering the high drag of the bracing wires of his rotor, and the aircraft did not fly. With a later version of his helicopter using a more powerful engine, some successful flights were accomplished, albeit under limited control. However, most flights resulted in damage or serious crashes followed by long periods of rebuilding. By 1925, Pescara had abandoned his helicopter projects.

Between 1924 and 1930, a Dutchman named A. G. von Baumhauer designed and built one of the first single-rotor helicopters with a tail rotor to counteract torque reaction. Boulet (1984) gives a good description of the machine. Figure 1.8 shows that the fuselage consisted essentially of a tubular truss, with an engine mounted on one end. The other end carried a smaller engine mounted at right angles to the main rotor shaft, which turned a conventional propeller to counter the main rotor torque reaction. The main rotor had two blades, which were restrained by cables so that the blades flapped about a hinge like a seesaw or teeter board. Control was achieved by a swashplate and cyclic-pitch mechanism, which was another very early application of this mechanism. Unfortunately, the main and tail rotors were in no way connected, and this caused considerable difficulties in achieving proper control. Nevertheless, the machine was reported to have made numerous short, semicontrolled flights.

In the late 1920s, the Austrian engineer Raoul Hafner designed and built a single-seat helicopter called the R-2 Revoplane – see Everett-Heath (1986) and Fey (1987). The flights were mostly unsuccessful despite some brief tethered flights of up to a minute. His early machines used a single-rotor configuration with a pair of fixed wings located in the rotor

downwash to provide antitorque. For rotor control, Hafner's machine is notable in that it used a swashplate for blade pitch, which was another early application of the mechanism. Hafner later emigrated to England, where he and Juan de la Cierva independently continued work on blade articulation for autogiros, and later, for helicopters.

1.3 The Era of the Autogiro

The Spanish engineer Juan de la Cierva had built and flown another type of rotating-wing aircraft as early as 1923 – see Juan de la Cierva (1926, 1930). This aircraft had a rotor that could freely turn on a vertical shaft but was not powered directly. Instead, the rotor disk was inclined backward at a small angle of attack, and as the machine was pulled forward by a propeller, the rotor was turned by the action of the airflow on the blades. This aerodynamic phenomenon, called autorotation, had been understood by Crocco and Yur'ev before 1910. Juan de la Cierva called his rotating-wing aircraft an "Autogiro." The name Autogiro was coined by de la Cierva as a proprietary name, but when spelled with a small "a" it is used as a generic name for this class of rotorcraft.

De la Cierva's first Autogiro was a coaxial design. The problem of asymmetric lift was well known to de la Cierva, and his idea of using a counter rotating coaxial design was that the lower rotor would counteract the asymmetry of lift produced on the upper rotor, thereby balancing the rolling moment on the aircraft. However, the aerodynamic interference produced between the rotors resulted in different rotor speeds, spoiling the required aerodynamic roll balance. De la Cierva conducted basic wind tunnel experiments on model rotors at Quatro Vientos near Madrid, and he was one of the first to establish a scientific understanding of their aerodynamic behavior. He built two more machines with single rotors before he achieved final success in January 1923 with the C-4. Based on his tests with small models, this machine incorporated blades with mechanical hinges at the root, which de la Cierva used as a means of equalizing the lift on the two sides of the rotor in forward flight – see de la Cierva & Rose (1931). The blades could flap up or down about these hinges, responding to the changing airloads during each blade revolution. Although the principle of flapping blades had actually been suggested by Charles Renard in 1904 and also patented by Louis Breguet in 1908, de la Cierva is credited with the first successful practical application.

In 1925, de la Cierva was invited to Britain by the Weir Company. His C-6 Autogiro was demonstrated at the Royal Aircraft Establishment (RAE), and these flights stimulated early theoretical work on rotating-wing aerodynamics at the RAE, mainly by Glauert and Lock. Early theoretical developments were also conducted by Munk in the United States. De la Cierva himself was to write two books for the fledgling rotorcraft industry, albeit formally unpublished, called *Engineering Theory of the Autogiro*, and *Theory of Stresses in Autogiro Rotor Blades*. In later models of his Autogiro, de la Cierva added a lag hinge to the blades, which alleviated stresses caused by in-plane Coriolis forces, and completed the development of the articulated rotor hub. A control stick was also connected to the rotor hub, which allowed the rotor disk to be tilted for control purposes (orientable direct rotor control). While this allowed the ailerons to be dispensed with, the rudder and elevator on the machine were retained. The Cierva Autogiro Company went on to build many more versions of the Autogiro, including the successful C-19 (Fig. 1.9), which is described by de la Cierva (1935).

Although the autogiro was still not a direct-lift machine and could not hover, it required only minimal forward airspeed to maintain flight. Juan de la Cierva proved that his autogiros were very safe and essentially stall-proof, and because of their low speed handling capability, they could be landed in confined areas. Takeoffs required a short runway, but this was



Figure 1.9 Cierva's C-19 successful two-seater autogiro, circa 1931.

rectified with the advent of the “jump” takeoff technique. To this end, Hafner introduced the “spider” cyclic-pitch control system to autogiros in 1934 – see Fay (1987). This provided a means of increasing collective pitch and also tilting the rotor disk without tilting the rotor shaft with a control stick as in de la Cierva’s direct control system. Hafner used this design in his third autogiro, the AR-3, which flew in 1935. In the “jump” takeoff technique, the blades are set to flat pitch and the rotor rpm is increased above the normal flight rpm using the engine. This is followed by the rapid application of collective pitch, while simultaneously declutching the rotor and thereby avoiding any torque reaction on the fuselage. This technique lifts the aircraft rapidly off the ground, powered only by the stored kinetic energy in the rotor system. As forward speed builds, the rotor settles into its normal autorotative state – see also Prewitt (1938). With its jump takeoff capability and low handling speeds, the autogiro was to closely rival the helicopter in performance capability.

Several other British companies including Weir, Avro, de Havilland, and Westland went on to build variants of the de la Cierva autogiro designs. The first Weir designs were developments of de la Cierva’s models and used the orientable direct rotor control system. The Weir W-1 through W-4 models were all autogiros and were some of the first machines to use a clutch to help bring up the rotor rpm prior to takeoff. The de Havilland and Westland companies built a few larger prototype autogiros. The Westland C-29 was a five-seat cabin autogiro built in 1934. The aircraft was never flown because it exhibited serious ground resonance problems, and the project was canceled with the untimely death of Juan de la Cierva in 1936. However, de la Cierva’s work was carried on by designers from Weir, and another Westland designed autogiro called the CL-20 was flown just before World War II – see Mondey (1982).

The Kellett and Pitcairn companies entered into licensing agreements with de la Cierva, resulting in the first flight of an autogiro in the United States in 1928. Pitcairn went on to design and patent many improvements into the de la Cierva rotor system [see Smith (1985)], but it became clear that it was a true helicopter with power delivered to the rotor shaft that was required. The autogiro was extensively tested in the United States by the NACA. Gustafson (1971) gives an authoritative account of the early NACA technical work on autogiros and

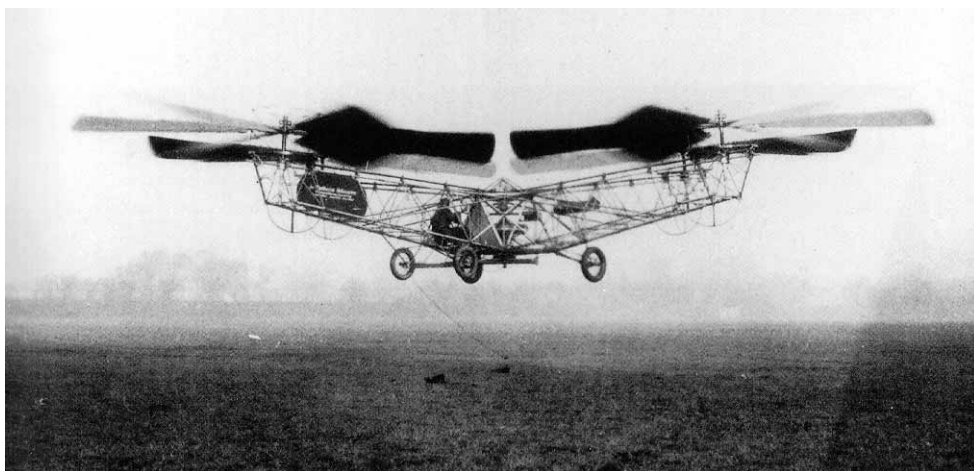


Figure 1.10 The Jerome–de Bothezat helicopter, which made limited controlled flights in 1922. (Courtesy NASM, Smithsonian Institution, SI Neg. No. 87-6022.)

helicopters.³ In Russia, the TsAGI built autogiros derived from the de la Cierva designs. Kuznetsov and Mil built the 2-EA, which was derived from the Cierva C-19 – see Everett-Heath (1988). Later developments of this design led to the first Russian helicopters built with the assistance of Vittorio Isacco, who had earlier led basic helicopter developments in Italy during the 1920s.

1.4 The First Successes with Helicopters

In 1922, a Russian émigré to the United States by the name of Georges de Bothezat built one of the largest helicopters of the time. De Bothezat had been a student of Professor Joukowski in Russia and had written one of the first technical manuscripts on rotating-wing aerodynamics – see de Bothezat (1919). De Bothezat’s machine was a quad-rotor with a rotor located at each end of a truss structure of intersecting beams, placed in the shape of a cross (Fig. 1.10). Ivan Jerome was the codesigner. Each rotor had six wide chord blades. Control of the machine was achieved by collective, differential collective, and cyclic blade pitch variations, and the design likely derived directly from those of Yur’ev. In 1922, the ungainly Jerome–de Bothezat quad-rotor flew successfully many times, albeit at low altitudes and forward speeds. However, because of insufficient performance and the increasing military interest in autogiros at the time, the project was canceled.

In 1920, Étienne Oemichen of France built a quad-rotor machine in a similar style to that of de Bothezat’s, but with a number of additional rotors for control and propulsion. His machine typified the cumbersome complexity of the various helicopters of the time. His initial design was underpowered, and it had to have a hydrogen balloon attached to provide additional lift and stability. However, he went on to design a “pure” helicopter that was flown between 1923 and 1924. By 1924, Oemichen’s machine proved “perfectly maneuverable and stable,” and he was awarded a prize by the FAI for demonstrating the first helicopter to fly a standard closed 1 km circuit. See also NACA (1921) and Oemichen (1923).

³ In addition, the entire first issue of the *Journal of the American Helicopter Society*, 1 (1), Jan. 1956, was devoted to the early autogiro and helicopter developments in the United States.

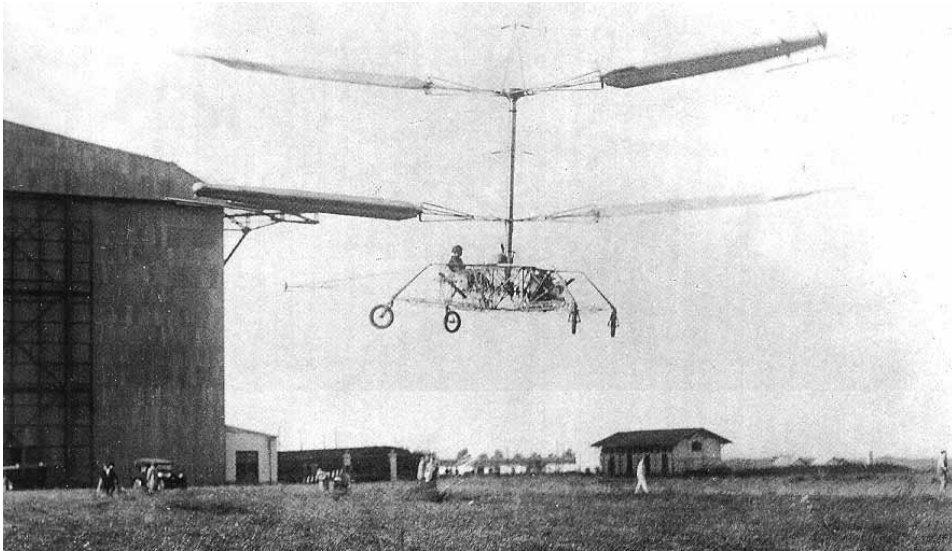


Figure 1.11 Corradino d'Ascanio's coaxial machine, circa 1930, which used servo-tabs on the blades to change their lift. (Courtesy John Schneider.)

In 1930, Corradino d'Ascanio of Italy built a highly successful coaxial helicopter. This machine held various speed and altitude records for the time. Figure 1.11 shows that this relatively large machine had two, two-bladed, counterrotating rotors. Following the work of de la Cierva, the blades had hinges that allowed for flapping and a feathering capability to change blade pitch. Control was achieved by auxiliary wings or servo-tabs on the trailing edges of the blades, a concept that was later adopted by others, including Bleeker and Kaman in the United States. D'Ascanio designed these servo-tabs so that they could be deflected periodically by a system of cables and pulleys, thereby cyclically changing the lift on the blade. For vertical flight, the tabs on all the blades moved collectively to increase the rotor thrust. Three small propellers mounted to the airframe were used for additional pitch, roll, and yaw control.

In 1930, Maitland Bleeker of the United States followed Brennan's approach to the torque reaction problem by delivering power to propellers that were mounted on each rotor blade. Power was supplied through a system of chains and gears from an engine mounted at the center of the machine. Like d'Ascanio's machine, Bleeker's helicopter was controlled by auxiliary aerodynamic surfaces he called "stabovators," which were fastened to the trailing edges of each blade. Both collective and cyclic pitch were incorporated into the design. Bleeker's machine accomplished numerous precarious hovers, but vibrations and control problems caused the project to be abandoned in 1933. Liberatore (1998) gives one of the best accounts of the project.

During 1929–1930, the Russian born engineer Nicolas Florine built a tandem rotor helicopter in Belgium. The rotors turned in the same direction but were tilted in opposite directions to cancel torque reaction. Boulet (1984) describes the mechanical aspects of the machine. Florine's first aircraft was destroyed in 1930, but he had a second design flying successfully by 1933, which made a flight of over 9 minutes and exceeded d'Ascanio's record of the time. Yet, Florine's designs suffered many setbacks, and work was discontinued in the mid 1930s.

During the period 1930–1936, the famous French aviation pioneers Louis Breguet and René Dorand made notable advances in helicopter development. Figure 1.12 shows their



Figure 1.12 The successful Breguet-Dorand coaxial helicopter, circa 1936. (Courtesy NASM, Smithsonian Institution, SI Neg. No. A-42078-c.)

machine of 1935, which was relatively large with coaxial rotors. Boulet (1984) and Kretz (1987) give an excellent account of the work. Each rotor had two tapered blades that were mounted with flap and lag hinges and were controlled in cyclic pitch using a swashplate design. Yaw control was achieved by differential torque on one rotor with respect to the other. Horizontal and vertical tails were used for increased stability. For its time, the aircraft had held several records, including a duration of 62 minutes and distance flown of 44 km (27 mi). Autorotations⁴ were attempted with the Breguet–Dorand machine, some moderately successful, but the aircraft crashed and further work stopped prior to the outbreak of World War II.

In 1937, Heinrich Focke in Germany, in conjunction with Georg Wulf and later Gerd Achgelis, demonstrated a successful lateral side-by-side, two-rotor machine, called the Fa-61. The details of the machine are described by Focke (1938, 1965) and Boulet (1984).

⁴ All aircraft must possess safe flight characteristics after a loss of power. While a fixed-wing aircraft can glide, the helicopter can take advantage of autorotation with the rotor unpowered as a means of maintaining rotor rpm, lift, and control in the event of engine failure. However, to get the rotor to autorotate, the helicopter must descend (at a relatively high rate) so that the relative wind comes upward through the rotor disk. The pilot, in effect, gives up altitude (potential energy) at a controlled rate for kinetic energy to drive the rotor and, with care, can “autorotate” the aircraft safely onto the ground. The ability to autorotate is a distinguishing feature of a successful helicopter.



Figure 1.13 The Focke Fa-61 flew successfully in 1938. (Courtesy NASM, Smithsonian Institution, SI Neg. No. A-2316.)

This machine was constructed from some rotor components provided by the Weir company. Figure 1.13 shows that the rotors were mounted on outriggers and were inclined slightly inward to provide lateral stability. The blades were tapered in planform and were attached to the rotor hub by both flapping and lagging hinges. Longitudinal control was achieved by tilting the rotors fore and aft by means of a swashplate mechanism, while yaw control was gained by tilting the rotors differentially. The rotors had no variable collective pitch, instead using a slow and clumsy system of changing rotor speed to change the rotor thrust. A vertical rudder and horizontal tail provided additional directional stability.

The Fa-61 machine is significant in that it was the first helicopter to demonstrate successful autorotations. Provision was made in the design for a fixed low collective pitch setting to keep the rotor from stalling during the descent. It also set records for duration (80 minutes), climb to altitude (3,427 m, 11,427 feet), forward speed (122 km/hr, 76 miles per hour), and distance (230 km, 143 miles). The machine gained a certain amount of notoriety prior to the outbreak of World War II when the famous test pilot Hanna Reitsch flew it inside a Berlin sports arena. The Fa-61 aircraft was used as a basis to develop the Fa-266 (Fa-233E), which first flew in 1940. This was a fairly large aircraft, with two three-bladed rotors, and could carry up to four crew. It went into limited production during the Second World War. Boulet (1984) gives a good account of the later helicopter work of Focke. After the war, some of the German machines were used as a basis to develop helicopters in Russia, for which a comprehensive account is given by Everett-Heath (1988).

With the assistance of Juan de la Cierva, the Weir Company had formed an aircraft department in Scotland in 1932. The W-5 was their first true helicopter design, since the W-1 through W-4 all had been autogiros. Initially, the W-5 was a coaxial design, but concerns about stability and control led to the redevelopment as a side-by-side configuration, which flew successfully in June 1938. Control was achieved with cyclic pitch but there was no collective pitch; vertical control was obtained by altering the rotor speed, a cumbersome feature used also on the Fa-61. The Weir W-5 (and later the W-6) and the Fa-61 were ahead of Sikorsky's VS-300. The W-6, which first flew in 1939, was a much larger version of the W-5 but still used the lateral side-by-side rotor configuration.

During the period 1938–1943, Antoine Flettner, also of Germany, developed several helicopter designs. Flettner's success came with using a side-by-side intermeshing rotor configuration, which became known as a *synchropter*. This rotor idea was first patented by Bourcart in 1903 and by Mees in 1910. In the synchropter design, the rotor shafts were close together but arranged so that they were at a significant outward angle with the overlapping rotors turning in opposite directions. A gearing system ensured the exact phasing of the rotors. In 1939, Flettner's Fl-265 synchropter was the first helicopter to demonstrate transition into autorotation and then back again into powered flight. Flettner built several other machines, including the Fl-282 Hummingbird.⁵ After World War II, in the United States, the Kellett Aircraft Company (which also built autogiros as a licensee to Pitcairn) adopted Flettner's synchropter configuration but used three-bladed instead of two-bladed intermeshing rotors. The aircraft flew very successfully, but it never went into production. The synchropter concept was also adopted by Charles Kaman, whose company Kaman Aircraft Corp. was later to put the type into successful production.

As described earlier, Igor Sikorsky had experimented in Czarist Russia with primitive vertical lift aircraft as early as 1907 – see Sikorsky (1938) and Finne (1987). After Sikorsky had emigrated to the United States, he went on to design and build giant flying boats. In 1935, Sikorsky was issued a patent, which showed a relatively modern looking single rotor/tail rotor helicopter design with flapping hinges and a form of cyclic pitch control. Although Sikorsky encountered many technical challenges, his first helicopter, the VS-300, was flying by May 1940. A good summary of the technical design is given by Sikorsky (1941, 1942, 1943). His first machine had one main rotor and three auxiliary tail rotors, with longitudinal and lateral control being obtained by means of pitch variations on the two vertically thrusting horizontal tail rotors. The main lifting rotor of the VS-300 was used in the later VS-300A, but only the vertical tail rotor was retained out of the original three auxiliary rotors for antitorque and directional (yaw) control purposes. In this configuration, longitudinal and lateral control was achieved by tilting the main rotor by means of cyclic-pitch control, while directional control was achieved by pitch variations of the tail rotor. This configuration was to become the standard for most modern helicopters. By 1941, Sikorsky had started production of the R-4 (Fig. 1.14) and in 1943 Sikorsky developed the R-5, which, although still only a two-seater helicopter, was much larger and had better performance than the R-4.

In 1946 Westland Helicopters in Great Britain obtained a license to build the Sikorsky machines, the first being designated as the WS-51 after the S-51, which was the first commercial helicopter designed by Sikorsky. This period was the start of a long relationship between the two companies which continues today. Westland already had a history as a successful fixed-wing manufacturer but decided to specialize in helicopters in 1946. After significantly reengineering the Sikorsky machine, Westland called the aircraft the Dragonfly. The Widgeon later followed, and this was a very modern looking and powerful version of the Dragonfly with a larger passenger cabin.

During 1942, the Weir Company, prompted by the success of Sikorsky's R-5, proposed a rather large single-rotor machine called the W-9, which was rather unique in its use of jet thrust to counteract rotor torque reaction – see Everett-Heath (1986). However, because the rotor lacked any collective pitch control, rotor thrust was controlled by changing rotor speed as in the pre-war Weir W-5/6 models. The W-9 crashed during a test flight in 1946, and the project was subsequently abandoned. The Weir and Cierva companies later went

⁵ The Fl-282 was probably one of the first helicopters in production; however, like the Focke helicopters, numbers were limited because of the Second World War.



Figure 1.14 The famous Sikorsky R-4B, circa 1944.

on to design the W-11 Air-Horse, which was an unorthodox three-rotor helicopter. The final helicopter of the Weir–Cierva line was the W-14 Skeeter, which was a small two-seater training helicopter, but this saw limited production.

It is significant to note that while helicopters were becoming more and more successful, the development of the autogiro continued in Europe and the United States well into the 1950s. Considerable development work was undertaken by The Pitcairn Company [see Pitcairn (1930) and Smith (1985)] and the Kellett Aircraft Companies in the United States. Harold Pitcairn patented over 100 concepts in rotor blade design and rotor control, some of which were later licensed to Sikorsky. The Pitcairn and Kellett autogiros or “Mailwings” flew on part of Eastern Airlines’ network delivering mail for the United States Post Office. Even into the 1960s small single- and two-seat autogiro designs were being developed in the United States by Umbaugh and McCulloch for the private market.

In Great Britain during the 1940s and 1950s, the autogiro concept was pursued to some significant end by the Fairey Aviation Company. The Fairey Girodyne compound aircraft used a propeller set on the end of a stub wing to provide both propulsion and antitorque – see Everett-Heath (1986). The Fairey Company went on to develop the Jet Girodyne in which the rotor system was driven by tip jets. This ultimately led to the Rotodyne, which was the world’s biggest giroplane with a cabin big enough for forty passengers – see Hislop (1958). The aircraft set a world speed record for a convertiplane in 1959 before the project was canceled. Smaller single- and two-seat autogiros were later built in Britain by the Wallis company.

Besides Igor Sikorsky, there were several other helicopter design pioneers in the United States during the 1940s. These included Arthur Young, Frank Piasecki, Stanley Hiller, and Charles Kaman. In the late 1930s, Arthur Young began a series of experiments with model helicopters that were ultimately to lead to the design of the renowned Bell-47 helicopter. After much research, Young invented the teetering rotor with a stabilizer bar; see Young (1948, 1979). The bar had bob weights attached to each end and was directly linked to the rotor blades through the pitch control linkages. The idea was that if the rotor was disturbed in pitch or roll, the gyroscopic inertia of the bar could be used to introduce cyclic pitch into the main rotor system, increasing the effective damping to disturbances and giving stability to the rotor system – see also Kelly (1954).



Figure 1.15 A version of the Bell-47, which was the world's first commercially certified helicopter. (By permission of Bell Helicopter Textron.)

The prototype Bell-30 was built in 1942 and had a single main teetering rotor with Young's stabilizer bar. The first untethered flights took place in 1943, and the machine was soon flying at speeds in excess of 70 mph. The Bell-30 led to the famous Bell Model 47 (Fig. 1.15), which was the world's first commercially certified helicopter. During its nearly thirty-year manufacturing period over 5,000 were produced in the United States alone, and many were also license built in more than twenty other countries. Tipton (1989), Brown (1995) and Spenser (1999) give good historical overviews of the Bell machines. Schneider (1995) gives a brief biography of Arthur Young and his novel teetering rotor design.

In 1943, Frank Piasecki designed and flew a small helicopter that was called the PV-2. This was the second successful helicopter to fly in the United States after Sikorsky's VS-300. Piasecki immediately turned to larger helicopters, and in 1943 the Piasecki Helicopter Corporation built a tandem rotor helicopter called the PV-3 Dogship. Details are given by the Piasecki Aircraft Corp. (1967) and Spenser (1999). This aircraft was popularly called the "Flying Banana" because of its distinctive curved fuselage shape. Despite its nickname, however, the aircraft was very successful and larger and more powerful versions quickly followed, including the H-16 and H-21 "Workhorse" models of 1952. The Vertol Company incorporated Piasecki's corporation in 1956 and went on to develop the Vertol 107 and two highly successful military tandem rotor models, the CH-46 and CH-47 (Fig. 1.16). The company finally became Boeing Helicopters. An overview of the Boeing-Vertol machines produced up to the mid-1970s is given by Grina (1975). In the late 1980s, the company produced a demonstrator of an advanced technology tandem rotor helicopter called the Model 360. The only other company in the United States to build a tandem helicopter design was Bell, who manufactured the XSL-1 during the 1950s. In 1998, Boeing announced the launch of the CH-47F and the CH-47SD "Super-D" Chinook. See also The Boeing Company (1999).

The British company, Bristol Helicopters, had designed and built a tandem helicopter during the late 1940s under the leadership of the helicopter pioneer Raoul Hafner – see Hobbs (1984) and Everett-Heath (1986). The Bristol Type-173 had a long, slim fuselage